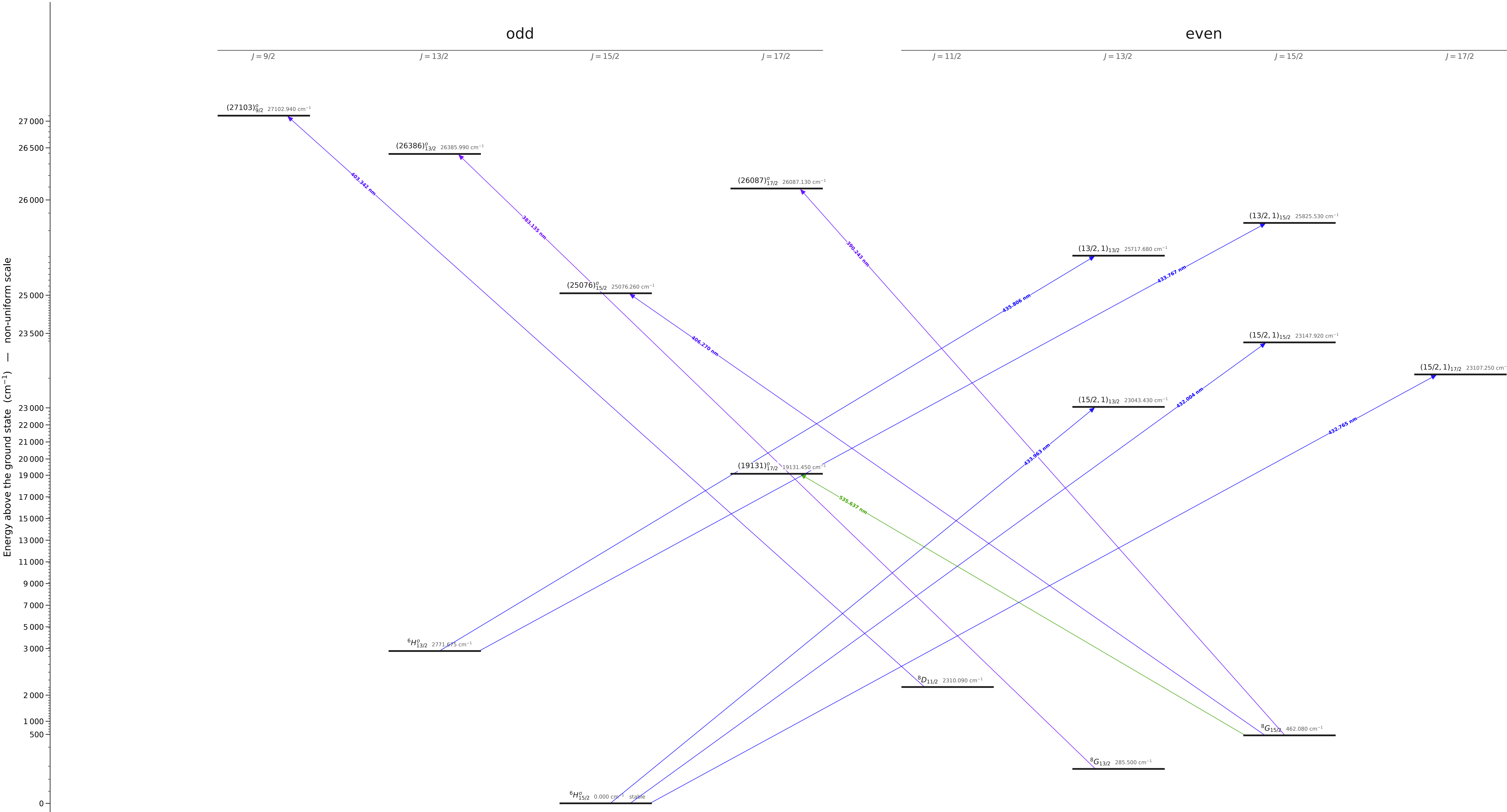
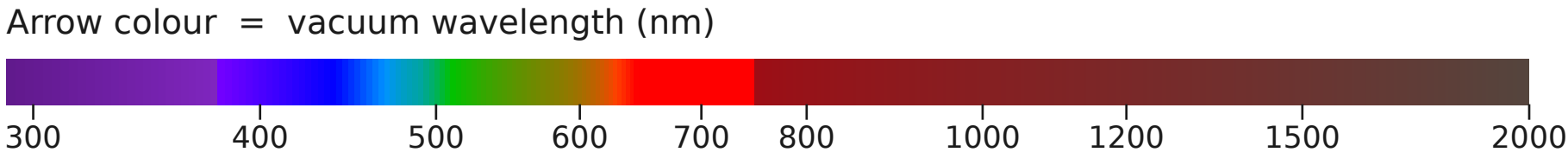


Terbium

Tb (Tb I). Strongest classified transitions below 2 μm (NIST): 15 levels, 10 lines.

Tb⁺ ionisation limit 47295.00 cm⁻¹ = 5.86383 eV (211.439 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J \| e r \| J' \rangle$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



¹⁵⁹Tb hyperfine structure (*I* = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$^8G_{13/2}$	531(6)	989(60)	F=5: -5588.86 F=6: -3013.88 F=7: +567.18 F=8: +5420.60
$^8G_{15/2}$	471(6)	1169(60)	F=6: -5575.57 F=7: -2982.07 F=8: +649.92 F=9: +5587.62

Measured A, B constants; sources in the data file.